

SMART INDIA HACKATHON 2026

TITLE PAGE

- Problem Statement ID – 26155
- Problem Statement Title – AI-Driven Multi-Vendor Network Security Compliance Auditor
- Theme – Blockchain & Cybersecurity
- PS Category – Software
- Team ID – 133478
- Team Name – Drone Syndicate

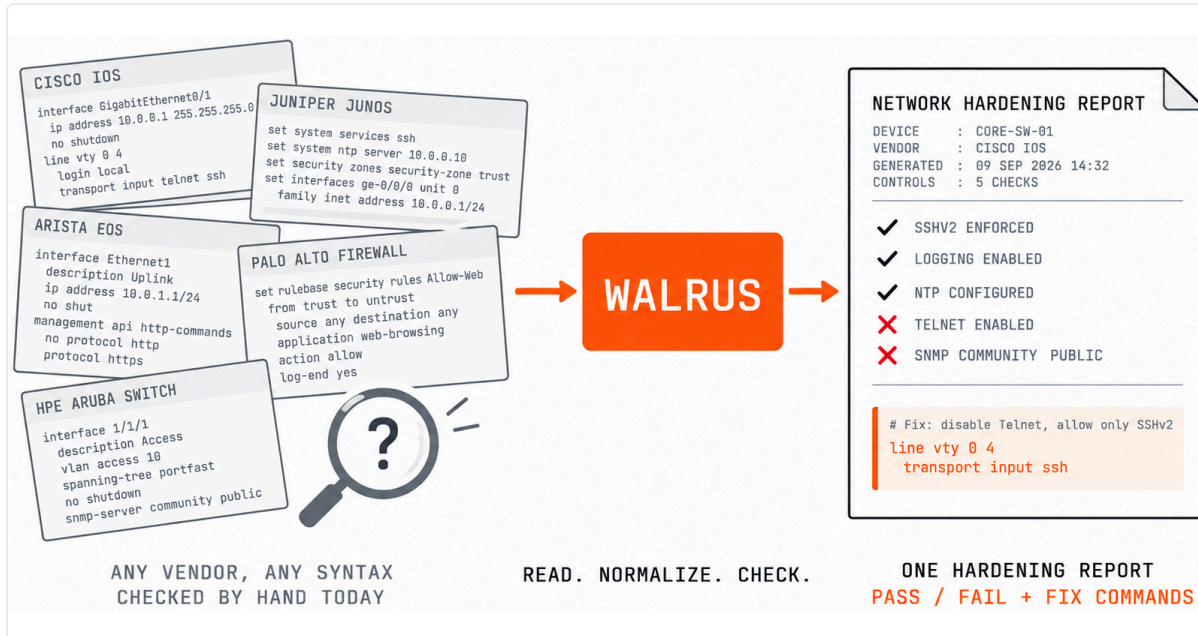
The network compliance auditor, a capability of the Walrus Securitas AI security platform



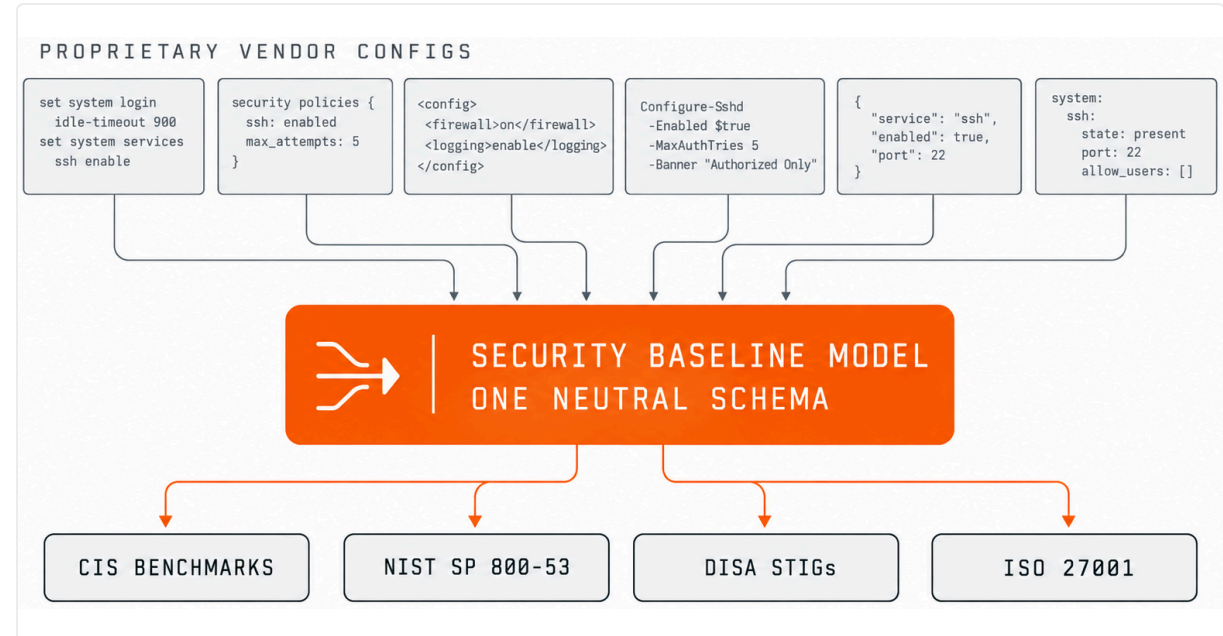
PROPOSED SOLUTION

Misconfiguration, not zero-days, causes ~99% of firewall breaches. Yet every device from 40+ vendors is still hardened by hand, one CLI at a time.

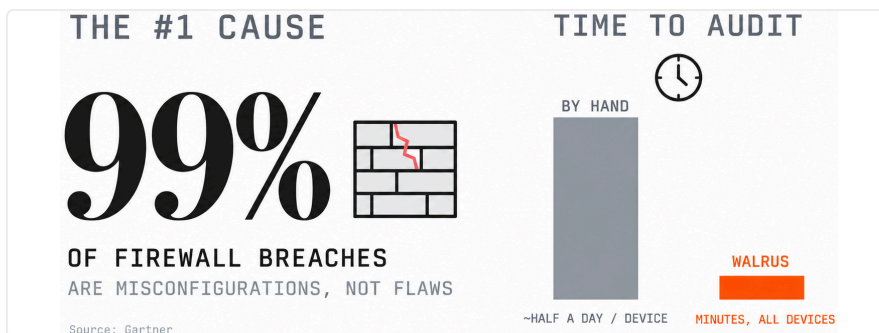
① ANY VENDOR CONFIG IN → ONE HARDENING REPORT OUT



② PROPRIETARY SYNTAX → ONE VENDOR-NEUTRAL MODEL



③ WHY IT MATTERS



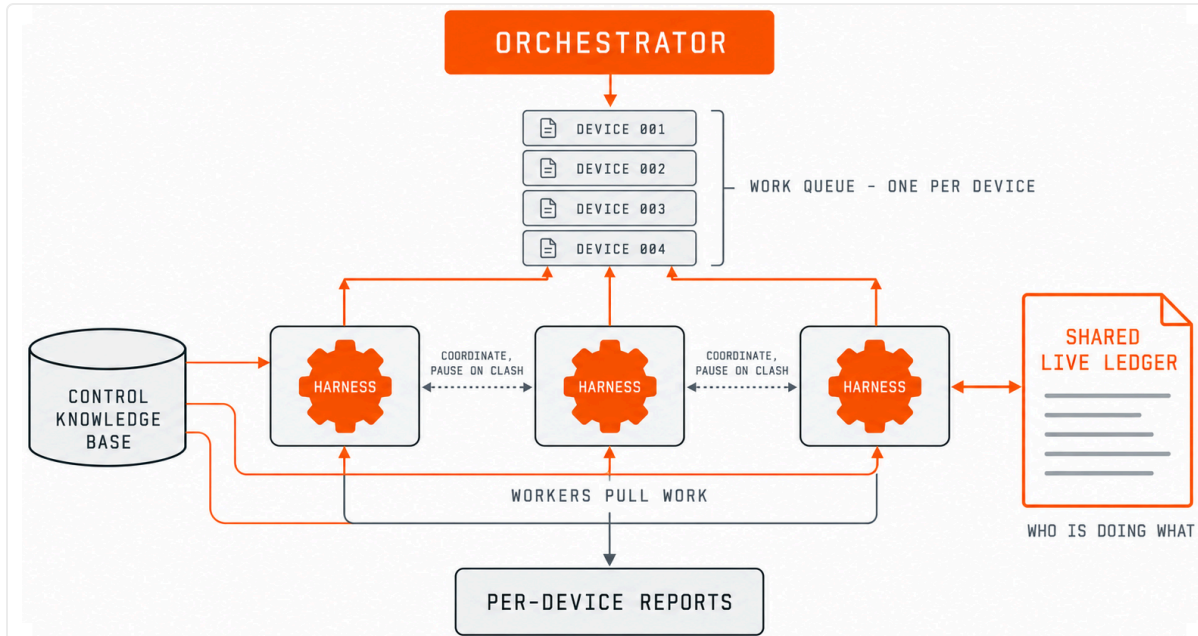
+Proposed Solution(Describe your Idea / Solution / Prototype)

- Detailed explanation:** Walrus ingests any network-device config from Cisco, Juniper, Arista, Palo Alto, firewalls, switches, cloud security groups. The AI reads it into **one vendor-neutral Security Baseline Model**; a deterministic engine then checks every control from CIS, NIST SP 800-53, DISA STIG and ISO 27001 and emits a per-device PDF with pass/fail, severity and the exact fix commands.
- How it addresses the problem:** It replaces slow manual checklist auditing and rigid, vendor-locked suites with **one source of truth** across a heterogeneous fleet. An unseen vendor is taught through a low-code training GUI, with **no code redeploy**.
- Innovation and uniqueness:** The AI **only parses**; the verdict is deterministic and every finding **cites its exact config line**, so it never hallucinates a pass/fail. A checklist gate makes it structurally **unable to skip a control**.

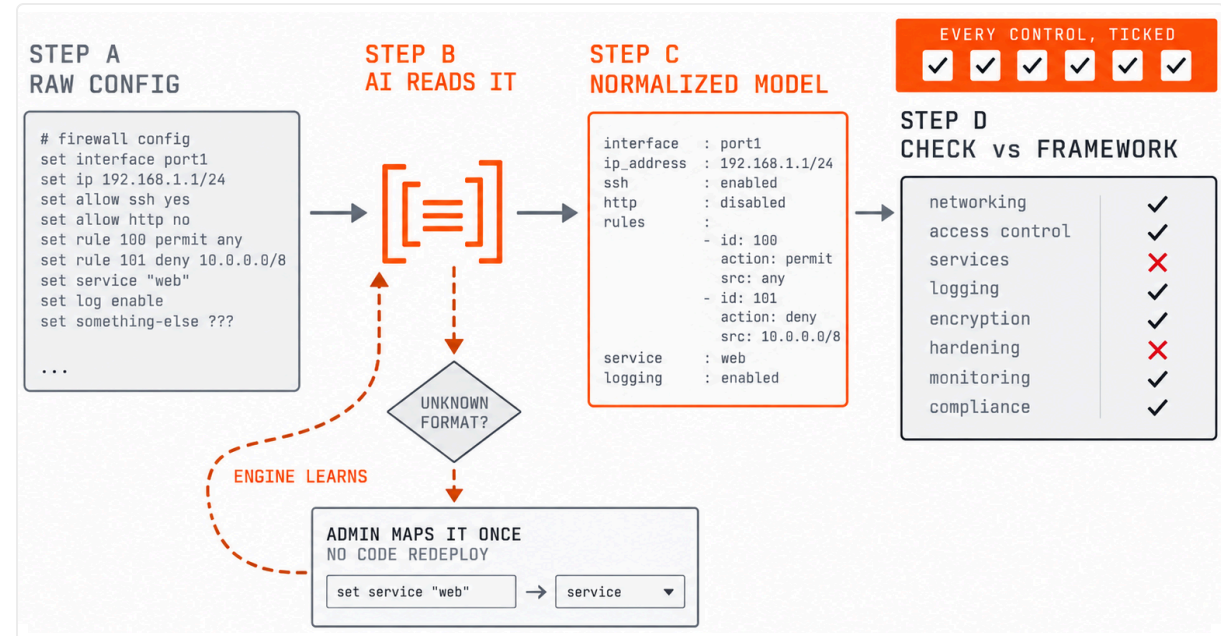
TECHNICAL APPROACH

Orchestrator → queue → a fleet of workers that pull work, each with its own harness and one shared live ledger, built so it cannot skip a single control.

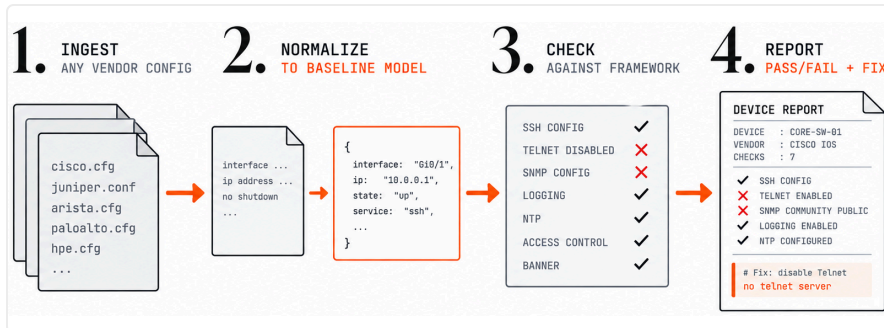
① THE ENGINE · WORKERS PULL, LEDGER TRACKS, NOTHING CLASHES



② THE AI PIPELINE · ONE CONFIG'S JOURNEY, WITH A LEARN LOOP



③ FOUR STAGES, END TO END

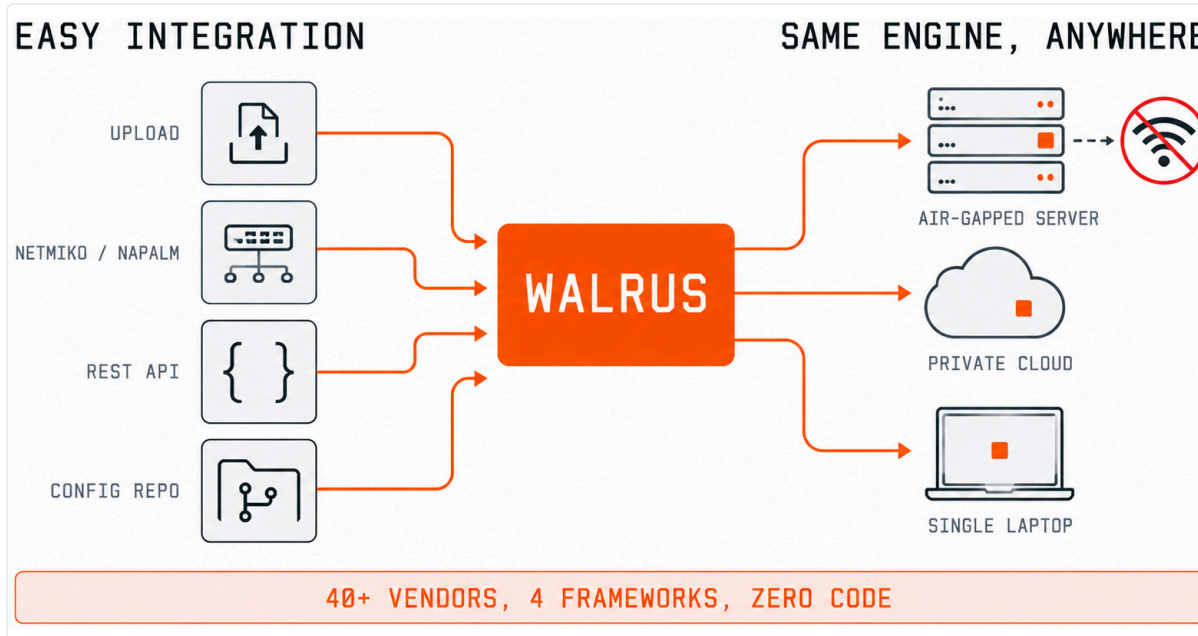


+Technical Approach

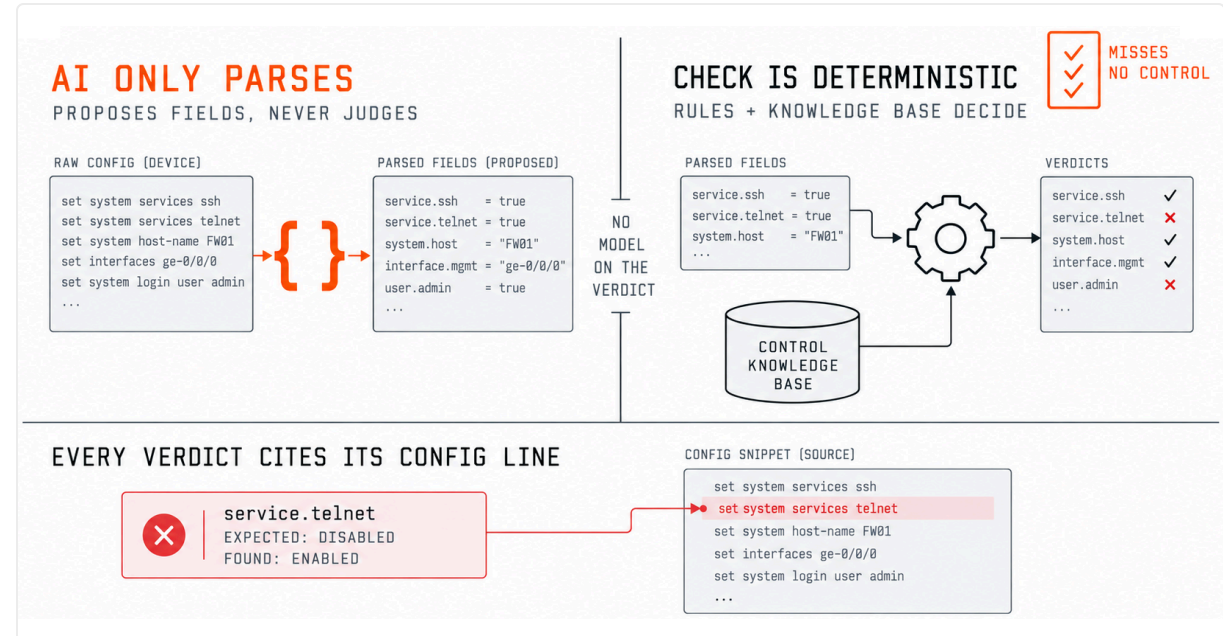
- **Technologies used:** Config collection via **Netmiko** / **NAPALM**, upload to a config repo; an **open-weight LLM** (NLP + pattern) for parsing, run on-prem; a deterministic rules engine over a **control knowledge base** (CIS / NIST 800-53 / STIG / ISO 27001); **ReportLab** for the per-device PDF. Built on the Walrus Harness engine.
- **Methodology and process:** The orchestrator writes one task per device to a queue; **workers pull work**, each running the harness, and a **shared live ledger** shows who is doing what and lets workers pause on a clash. When a config cross-references another object such as a nested ACL or a linked device, it is re-queued and run through the **full control set**, so coverage never leaks.
- **Never miss, never lie:** The knowledge base is the checklist. The fleet must return a verdict for every applicable control before an audit can close; the model only proposes fields, the engine decides, and every verdict is backed by an **append-only evidence log**.

Built on proven pieces like the Walrus Harness engine, Netmiko/NAPALM and public benchmarks, and it runs air-gapped, so it fits critical infrastructure.

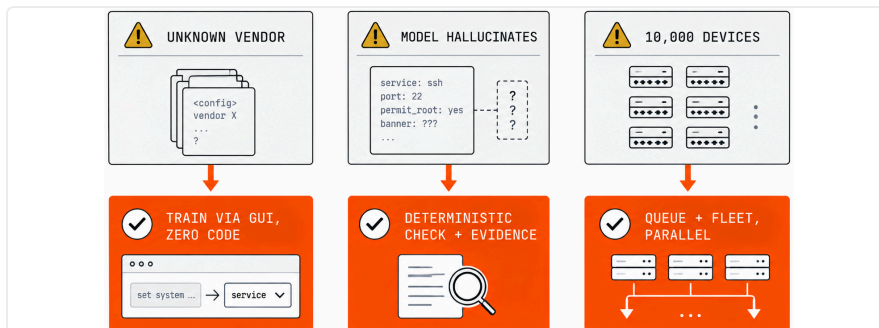
① RUNS ANYWHERE · EASY TO INTEGRATE



② WHY IT CANNOT HALLUCINATE A VERDICT



③ THREE RISKS, THREE ANSWERS



+Feasibility and Viability

- Feasibility:** The orchestration, evidence and checklist engine already exists (Walrus Harness). Netmiko/NAPALM and the four frameworks are mature and public; an open-weight model runs on a mid-range GPU on-prem. Deployable on an **air-gapped server, private cloud or a single laptop**.
- Potential challenges and risks:** Unseen vendor firmware breaks fixed parsers · an LLM can hallucinate · a 10,000-device fleet is slow to audit by hand · configs cross-reference each other.
- Strategies for overcoming them:** Unknown vendor → **train via GUI, zero code**. Hallucination → **deterministic check + cited evidence**, no model on the verdict. Scale → **queue + worker fleet, in parallel**. Cross-references → re-queued through the full control set. And **nothing leaves the building** with no external calls, fit for NCIIPC-class infrastructure.

IMPACT AND BENEFITS

A manual audit of one device takes an analyst half a day and still skips controls under time pressure. Walrus does the whole fleet in minutes, missing none.

① DAYS → MINUTES · 100% OF CONTROLS, EVERY RUN

TIME TO FULL AUDIT

MANUAL - DAYS

10

WALRUS - MINUTES

0.2

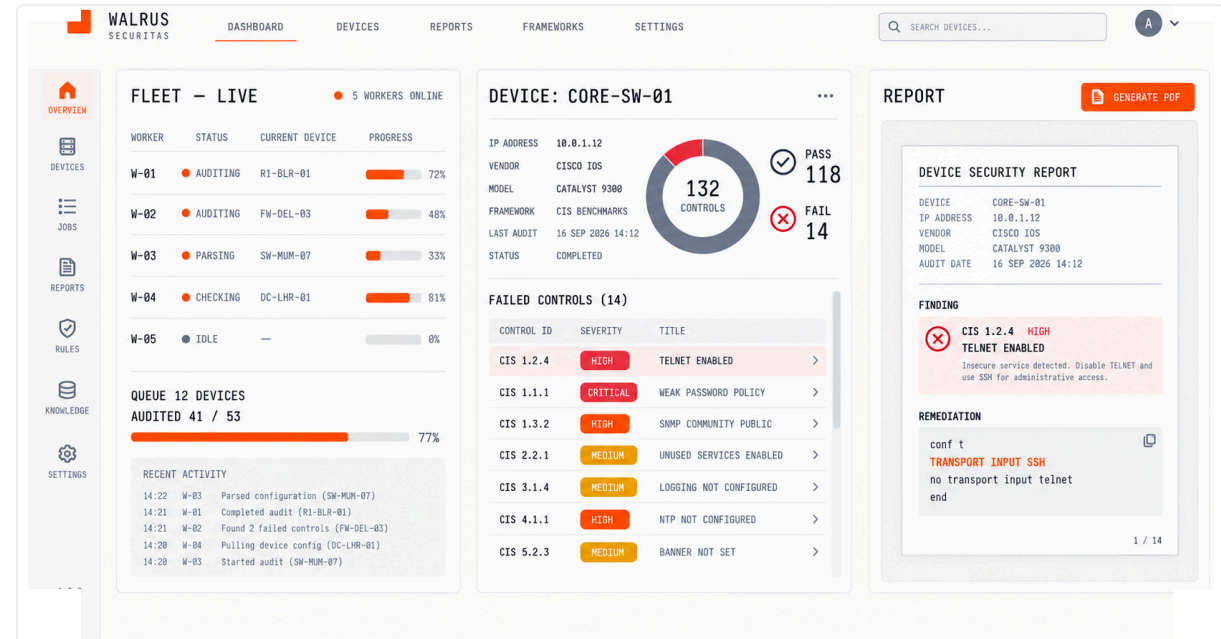
CONTROL COVERAGE

100%

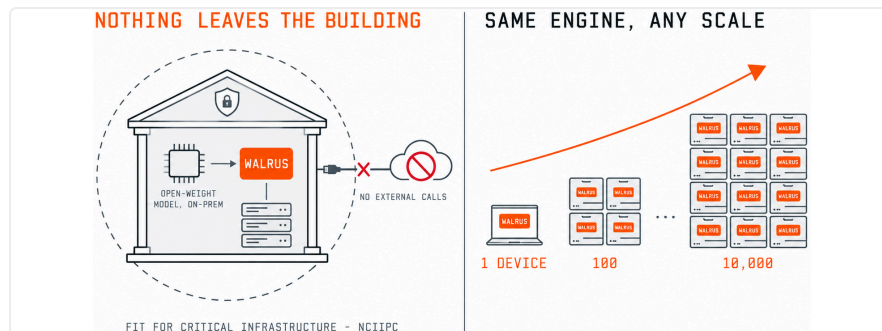
MANUAL AUDITS
SKIP UNDER TIME

OF APPLICABLE CONTROLS, EVERY RUN

② WHAT THE OPERATOR SEES · LIVE FLEET, PER-DEVICE REPORT



③ SOVEREIGN, AND IT SCALES



+Impact and Benefits

- Impact on the target audience:** NOC and security teams, CISOs and **critical-infrastructure operators (NCIIPC)** get a continuous, single source of truth for hardening compliance across **any** vendor, so the audit that took days now runs before the change window closes.
- Security:** Misconfiguration is the **#1 cause of breaches**; catching every hardening gap on every device shrinks exactly the attack surface that causes most of them.
- Economic & environmental:** Replaces expensive vendor-locked suites and days of analyst time; **one engine scales 1 → 10,000 devices** with no per-vendor licensing. Open-weight and on-prem: no cloud spend, **no data leaves the building**.

Every control traces to a published standard, and every verdict in the tool cites the exact config line it came from.

+Research and References

FRAMEWORKS & STANDARDS

- ▶ **CIS Benchmarks:** per-vendor hardening baselines (Cisco, Juniper, Arista, PAN-OS...)
- ▶ **NIST SP 800-53 Rev. 5:** security & privacy controls catalogue
- ▶ **DISA STIGs:** hardening guides, per network device
- ▶ **ISO/IEC 27001:2022:** Annex A, 93 controls
- ▶ **NCIIPC:** Critical Information Infrastructure protection (PS reference)

METHOD & TOOLING

- ▶ **Netmiko / NAPALM:** multi-vendor config collection
- ▶ **Open-weight LLM:** on-prem parsing, no external calls
- ▶ **ReportLab:** per-device PDF with remediation CLI
- ▶ **Walrus Harness:** our engine, evidence log + the 'miss-nothing' checklist gate
- ▶ **Gartner:** ~99% of firewall breaches are misconfigurations, not flaws

OUR OWN BUILD

- ▶ **Live product page + demo:** open the link below or scan the code
- ▶ **Runs air-gapped:** open-weight model, on-prem, no data leaves
- ▶ **Deterministic verdicts:** every finding cites its config line
- ▶ Part of the **Walrus Securitas** AI security platform



+ TRY THE LIVE NETWORK COMPLIANCE AUDITOR

walrussecuritas.com/product/network-compliance-auditor

Scan the code or open the link for a real audit on sample Cisco / Juniper / Arista configs, in your browser.

FRAMEWORK COVERAGE · ONE NEUTRAL MODEL, FOUR STANDARDS, ANY VENDOR

FRAMEWORK	WHAT IT HARDENS	EXAMPLE CONTROL THE ENGINE CHECKS	IN SCOPE
CIS Benchmarks	Per-vendor device baselines	SSH v2 enforced; Telnet / HTTP disabled	Cisco, Juniper, Arista, PAN-OS, ...
NIST SP 800-53	Access, audit, comms controls	AC-17 remote access; AU-12 audit logging	All device classes
DISA STIG	DoD hardening, per device	SNMP community not 'public'; NTP set	Routers, switches, firewalls
ISO/IEC 27001	Annex A, 93 controls	A.8.20 network security; A.8.9 config mgmt	Org-wide baseline
New / white-box vendor	Unrecognised CLI	taught once via the low-code training GUI, then checked against all of the above	